

The Maximum Number of Vertices of a ($d, d + 1$)-Zonoboxtope Is $2^{d+2} - 6$

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Abstract

A (d, n) -zonoboxtope is the convex hull of two weighted Minkowski sums of a common family of n line segments in $\mathbb{R}^d$. We determine its maximal vertex number when $n = d + 1$. For every $d \geq 2$, the maximum is

$$2^{d+2} - 6.$$

The proof reduces the separating-hypersurface test to comparisons between two lists of $d + 1$ real numbers. Their global maximum and global minimum force two distinct chambers to be monocolored, giving the upper bound. An explicit rational family has exactly those two monocolored chambers and attains the bound. Consequently, the $(4, 5)$ maximum is 58, and Part 2 of Conjecture 6.6 in Balakin–Cox–Loho–Sturmfels is false for every $d \geq 4$ at $n = d + 1$.

1 Statement and context

Write each segment as

$$I_i = m_i + [-u_i, u_i] \subset \mathbb{R}^d,$$

and let

$$Z_a = \sum_{i=1}^n a_i I_i, \quad Z_b = \sum_{i=1}^n b_i I_i, \quad Q = \text{conv}(Z_a \cup Z_b), \quad (1)$$

where $a_i, b_i \geq 0$. The polytope Q is a (d, n) -zonoboxtope.

Balakin, Cox, Loho, and Sturmfels conjectured that, for $4 \leq d \leq n$, the maximal number of vertices is

$$4 \sum_{k=0}^{d-1} \binom{n-1}{k}. \quad (2)$$

At $n = d + 1$, this equals $2^{d+2} - 4$.

Theorem 1. *For every integer $d \geq 2$, the maximum number of vertices of a $(d, d + 1)$ -zonoboxtope is*

$$\boxed{2^{d+2} - 6}.$$

The same maximum holds for zonoboxtope candidates, in which the relative translation of the two zonotopes is unrestricted.

The values for the first dimensions are shown below.

d	$n = d + 1$	formula (2)	exact maximum
2	3	12	10
3	4	28	26
4	5	60	58
5	6	124	122
6	7	252	250

The first two exact values were already known from the two- and three-dimensional results in [1]. The theorem supplies the general pattern and, starting at $d = 4$, an infinite family of counterexamples to (2).

2 The chamber reduction

It is enough to prove the result in generic position. Indeed, the vertices of the convex hull of finitely many continuously varying candidate points persist under sufficiently small perturbations: for each old vertex, an exposing functional has a positive gap from all distinct competing points. Thus the parameters in (1) may be perturbed so that the u_i are in general linear position, all a_i, b_i are positive, and all comparisons below are strict, without decreasing the vertex number.

Set $n = d + 1$. There is a unique linear dependence, up to scale,

$$\sum_{i=1}^n c_i u_i = 0, \quad (3)$$

and every c_i is nonzero. Replacing u_i by $-u_i$ leaves I_i unchanged, so we may assume $c_i > 0$ for every i .

For $v \in \mathbb{R}^d$, put $y_i = \langle u_i, v \rangle$. The map $v \mapsto y$ identifies $\mathbb{R}^d$ with the hyperplane

$$H = \left\{ y \in \mathbb{R}^n : \sum_{i=1}^n c_i y_i = 0 \right\}.$$

The chambers of the common normal fan of Z_a and Z_b are indexed by the nonempty proper subsets $P \subset [n]$:

$$C_P = \{y \in H : y_i > 0 \ (i \in P), \ y_j < 0 \ (j \notin P)\}.$$

Hence their number is

$$R = 2^n - 2 = 2^{d+1} - 2. \quad (4)$$

In the coordinates $z_i = c_i y_i$, the closure of C_P is generated by the rays

$$e_i - e_j, \quad i \in P, \ j \notin P. \quad (5)$$

This follows, for example, by viewing a vector with positive coordinates on P , negative coordinates off P , and zero coordinate sum as a nonnegative transportation flow from P to its complement.

The difference of the two support functions is

$$g(v) = h_{Z_a}(v) - h_{Z_b}(v) = \langle T, v \rangle + \sum_{i=1}^n \delta_i |y_i|, \quad T = \sum_{i=1}^n \delta_i m_i, \quad \delta_i = a_i - b_i. \quad (6)$$

Choose $t \in \mathbb{R}^n$ with $U^T t = T$, where the rows of U are the u_i . Define

$$p_i = \frac{t_i + \delta_i}{c_i}, \quad q_i = \frac{t_i - \delta_i}{c_i}. \quad (7)$$

On the ray (5), an immediate substitution into (6) gives

$$g(e_i - e_j) = p_i - q_j. \quad (8)$$

Here and below, $g(e_i - e_j)$ denotes the value on the corresponding ray after the z -coordinate normalization.

Call C_P *bicolored* if the set

$$\{p_i - q_j : i \in P, j \notin P\} \quad (9)$$

contains both positive and negative values; otherwise call it *monocolored*. The separating-hypersurface formula of [1, Proposition 6.3] becomes

$$f_0(Q) = R + \#\{P : C_P \text{ is bicolored}\}. \quad (10)$$

For completeness, the reason is visible directly from (8): in a bicolored chamber the hyperplane $g = 0$ cuts the chamber and both corresponding zonotope vertices survive in Q ; in a monocolored chamber only the vertex from the dominating zonotope survives.

3 Two chambers must be monocolored

Lemma 2. *Assume $n \geq 3$ and the $2n$ numbers $p_1, \dots, p_n, q_1, \dots, q_n$ are distinct. Then at least two distinct chambers C_P are monocolored.*

Proof. Consider the global maximum among the $2n$ numbers. If it is p_r , then every value $p_r - q_j$ with $j \neq r$ is positive, so $C_{\{r\}}$ is monocolored. If it is q_r , then every value $p_i - q_r$ with $i \neq r$ is negative, so $C_{[n] \setminus \{r\}}$ is monocolored.

The global minimum gives another monocolored chamber by the same argument with signs reversed: p_s gives $C_{\{s\}}$, while q_s gives $C_{[n] \setminus \{s\}}$. These two chambers are distinct. For extrema of the same type this follows from distinctness; for extrema of opposite types it follows because a singleton cannot equal the complement of a singleton when $n \geq 3$. $\square$

Combining Lemma 2 with (10) gives

$$f_0(Q) \leq R + (R - 2) = 2R - 2 = 2^{d+2} - 6.$$

The perturbation observation at the start of Section 2 extends this bound to all, possibly nongeneric, instances.

4 A rational family attaining the bound

We now construct an instance with exactly two monocolored chambers in every dimension. Use zero-based indices $i = 0, \dots, n - 1$, with $n = d + 1$, and set

$$u_i = e_{i+1} \quad (0 \leq i \leq n - 2), \quad u_{n-1} = -(1, \dots, 1). \quad (11)$$

Every d of these vectors are independent, and their positive circuit has $c_i = 1$ for all i .

Prescribe the ray-comparison parameters

$$p_i = 2i \quad (0 \leq i \leq n - 1), \quad q_i = \begin{cases} 2i + 3, & 0 \leq i \leq n - 2, \\ 1, & i = n - 1. \end{cases} \quad (12)$$

These are realized by rational zonoboxtope data as follows:

$$t_i = \frac{p_i + q_i}{2}, \quad \delta_i = \frac{p_i - q_i}{2}, \quad b_i = 2n, \quad a_i = 2n + \delta_i. \quad (13)$$

All weights are positive. Let $T = U^\top t$. Since $\delta_0 = -3/2$, take

$$m_0 = \frac{T}{\delta_0} = -\frac{2}{3}T, \quad m_i = 0 \quad (i > 0), \quad (14)$$

and define $I_i = m_i + [-u_i, u_i]$. Then $\sum_i \delta_i m_i = T$, so (7) recovers exactly (12). Explicitly, the coordinates of T are

$$T_k = 2k + 2 - n, \quad k = 0, \dots, d - 1.$$

It remains to count the monocolored chambers. A chamber is negative-monocolored exactly when

$$\max_{i \in P} p_i < \min_{j \notin P} q_j. \quad (15)$$

If $n - 1 \notin P$, then $q_{n-1} = 1$ occurs on the right of (15); hence (15) forces $P = \{0\}$. If $n - 1 \in P$, then the left side is $p_{n-1} = 2n - 2$; the only q_j larger than this is $q_{n-2} = 2n - 1$, so (15) forces $P = [n] \setminus \{n - 2\}$. Both of these chambers are indeed negative-monocolored.

There are no positive-monocolored chambers. If $s = \min P = 0$, then $p_s = 0$ cannot exceed the positive q_j off P . If $s \geq 1$, then $s - 1 \notin P$ and

$$p_s = 2s < 2s + 1 = q_{s-1},$$

again preventing all cross-differences from being positive. Thus exactly two of the R chambers are monocolored. Equation (10) now gives

$$f_0(Q) = R + (R - 2) = 2^{d+2} - 6,$$

which proves the lower bound and completes the proof of Theorem 1.

5 Consequences and verification

Corollary 3. *The maximal number of vertices of a $(4, 5)$ -zonoboxtope is 58.*

Corollary 4. *Part 2 of Conjecture 6.6 in [1] is false for every $d \geq 4$: at $n = d + 1$ it predicts $2^{d+2} - 4$, while the exact value is $2^{d+2} - 6$.*

The accompanying script `verify_d_dplus1.py` uses only the Python standard library. For any requested finite range of dimensions it checks the rational realization, the unique positive circuit, all chamber signs, the two exceptional subsets, and the resulting vertex count. The script is supplementary: the proof above is symbolic and applies to every $d \geq 2$.

Status and provenance. This is a self-contained research draft and has not yet undergone peer review. The reduction, extremal argument, and general construction were developed on August 2, 2026 in an AI-assisted research session directed by the author using ChatGPT (GPT-5.6). The mathematical claims rest on the proof above and the independently runnable exact checker, not on unverified model output.

References

- [1] A. Balakin, S. Cox, G. Loho, and B. Sturmfels, *Maxout Polytopes*, arXiv:2509.21286, 2025. <https://arxiv.org/abs/2509.21286>